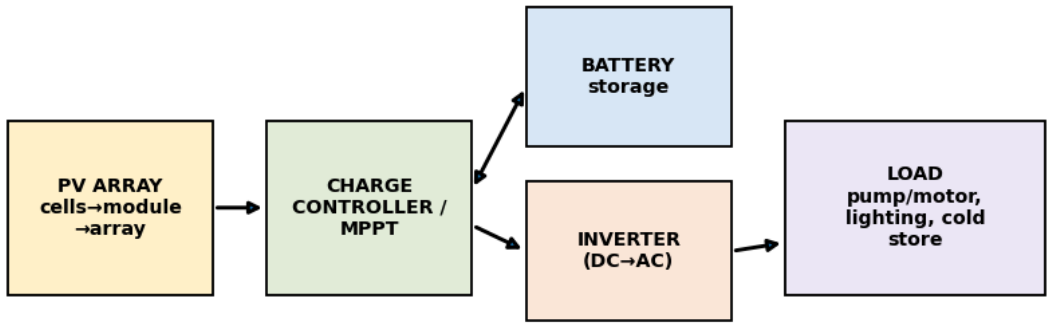
UNIT 7 · TOPIC 4(B)

SPV-Powered Systems in Agriculture

Syllabus topic: Solar photovoltaic conversion and SPV powered systems

The system chain



Solar photovoltaic (SPV) system and storage options

1. PV array.
2. Charge controller / MPPT.
3. Battery (optional).
4. Inverter for AC loads (optional).
5. Load.

Storage choices

- **Batteries** (lead-acid or Li-ion — Unit 3, Topic 12c) for lighting and electronics.
- **Store the OUTPUT instead:** pump water to a TANK — the battery-less solar pump is the cheapest and the dominant PM-KUSUM design; or use thermal mass, or ice/cold for cold storage.

Solar pump — the flagship on-farm SPV application

1. PV array.
2. MPPT / variable-frequency drive (VFD).
3. DC or AC motor–pump (submersible or surface).

- Size by dynamic head and daily water need.
- No fuel, low O&M (PM-KUSUM Components B and C — Topic 1).

Other SPV uses

- Solar lighting and fencing.

- PV-powered sprayers/dusters and small tools.
- Cold storage and milk chillers; dryers with PV fans.
- Weather stations and IoT sensors (Unit 2, Topic 10).
- Grid-tied rooftop systems (PM Surya Ghar).

> Design rule: match array output to the load profile and, where storage is limited, to the daily solar window.

Practice MCQs

Q67. The cheapest storage for a solar irrigation pump is usually to:

(a) Use large batteries
(b) Pump water into a tank (store water, not charge)
(c) Use PCM
(d) Use a flywheel

Q68. A solar pump system typically comprises:

(a) Array → MPPT/VFD → motor–pump
(b) Array → boiler
(c) Array → engine
(d) Battery → diesel

Q69. An inverter in an SPV system:

(a) Stores energy
(b) Converts DC to AC for AC loads
(c) Tracks the sun
(d) Cools cells

Q70. A major practical loss for field PV in India is:

(a) Too much diffuse light
(b) Dust deposition and shading (needs cleaning)
(c) Excess voltage
(d) Low band gap

Answers

Q67 — (b) Pump water into a tank (store water, not charge) — *Battery-less solar pumps store WATER in a tank (cheapest) — the PM-KUSUM standard.*

Q68 — (a) Array → MPPT/VFD → motor–pump — *Chain: array → MPPT/VFD → motor–pump.*

Q69 — (b) Converts DC to AC for AC loads

Q70 — (b) Dust deposition and shading (needs cleaning)

Sources: Sukhatme & Nayak — Solar Energy; Duffie & Beckman — Solar Engineering of Thermal Processes; Rai — Non-Conventional Energy Sources; NREL/ITRPV and manufacturer datasheets (efficiency figures as given in the source notes, July 2026); MNRE guidelines; NPTEL solar-energy lectures.